TECHNICAL ARCHITECTURE & STACK SPECIFICATION

University Panic Button System (UPBS)

End-to-End Technology Stack, Component Allocation & Architectural Justification

Project: University Panic Button System (UPBS)
Institution: Sol Plaatje University (Kimberley, SA)
Production URL: https://youthalliance.co.za
Hosting Environment: Render Linux Cloud PaaS
Database Engine: Supabase Managed Cloud PostgreSQL 15.x
Academic Code: NPRT630 Final Software Engineering Capstone

Architectural Design Paradigm: UPBS is engineered as a high-reliability *hybrid emergency dispatch platform* combining a modular **Django 5.x / Python 3.12+ backend core** with an ultra-responsive glassmorphic web application and a native Android companion client. It provides sub-second emergency dispatch CAD, offline-resilient Leaflet GIS mapping, biometric & GPS telemetry acquisition, and cryptographic RBAC isolation.

1. Core Backend & Web Application Layer

Server-Side Runtime

TECHNOLOGY	VERSION & SCOPE	ROLE & PURPOSE IN UPBS ARCHITECTURE
Python `v3.12+ Runtime`	Core Programming Language	Powers the complete server-side ecosystem. Handles business logic, Haversine spatial telemetry calculations, reverse geocoding heuristics, API endpoints, authentication flows, and background maintenance automation.
Django `v4.2.x LTS / v5.x`	Full-Stack Web Framework	Provides the MTV (Model-Template-View) backbone, Object-Relational Mapper (ORM), URL routing, declarative forms, session management, and enterprise-grade security middleware (CSRF protection, XSS escaping, SQL injection prevention, clickjacking defense).
Gunicorn `v21.2.0`	WSGI Production HTTP Server	Acts as the production UNIX Web Server Gateway Interface. Runs multi-worker pre-fork processes on Render to handle concurrent requests for live GPS pings, dispatch broadcasts, and web app assets with zero downtime.
WhiteNoise `v6.6.0`	Static File Asset Server	Serves pre-compressed (gzip/Brotli) CSS, JavaScript, icons, and offline Kimberley GIS map tiles directly out of the Gunicorn container with immutable caching headers, eliminating the need for Nginx or AWS S3 buckets.
Pillow `v10.0.0`	Python Imaging Library	Handles photo uploads for campus hazard and crime incident reporting. Validates file integrity, sanitizes EXIF metadata for privacy, resizes images, and manages image conversions safely.
Requests `v2.31.0`	Synchronous HTTP Client	Executes external REST API requests for transactional email delivery via Resend, upstream Nominatim reverse geocoding, and client IP geolocation resolution.

2. Database & Data Persistence Infrastructure

ACID Relational Storage

TECHNOLOGY	CONFIGURATION	ROLE & PURPOSE IN UPBS ARCHITECTURE
Supabase PostgreSQL `PostgreSQL 15.x`	Production Relational DB (Cloud)	Stores all production institutional principals, custom User models, RBAC credentials, emergency distress incidents, GPS breadcrumbs, officer CAD dispatches, hazard reports, and ISO usability feedback with full ACID guarantees.
SQLite 3 `v3.x Embedded`	Local Development & CI Testing	Provides lightweight, zero-configuration local persistence during development, offline evaluation, and rapid automated test execution (executes all 43 unit and integration tests in ~19 seconds).
dj-database-url `v2.1.0`	Dynamic Connection Parser	Enables seamless 12-factor app configuration by dynamically reading the **DATABASE_URL** environment variable to toggle between SQLite locally and Supabase PostgreSQL in production without code edits.
psycopg2-binary `v2.9.9`	High-Speed DB Driver	C-optimized PostgreSQL database adapter connecting Django's ORM to Supabase with persistent connection pooling (**conn_max_age=600**) and health checks.

3. GIS, Geolocation & Computer-Aided Dispatch (CAD) Spatial Emergency Engine

GPS HTML5 Geolocation API

Direct browser and native hardware interface utilizing `enableHighAccuracy: true`, satellite GPS, Wi-Fi BSSID triangulation, and cellular cell towers to lock student coordinates with sub-5-meter precision.

GIS Leaflet.js Mapping Engine (v1.9.4)

Mobile-optimized, zero-dependency spatial mapping library powering the Control Room Tactical Radar, student location preview, patrol officer tracking, and campus landmark markers.

MAP OpenStreetMap & CARTO Dark Matter

High-contrast, inverted tactical dark-mode base map tiles providing clear night-vision visualization for campus security dispatchers with zero light pollution.

CACHE Kimberley Tile Cache (Local Offline)

Over 130 offline map tiles pre-cached in `static/tiles/kimberley/` ensuring CAD map visualization remains functional even during full internet outages or tile server rate limits.

COMPONENT	IMPLEMENTATION	ROLE & MATHEMATICAL / TECHNICAL PURPOSE
Haversine Geodesic Engine	Pure Python Algorithm (**math.atan2**)	Calculates great-circle distance between student distress coordinates and all patrol officers on duty to dynamically rank and assign the closest responder in meters.
Forward Compass Bearing	Trigonometric Vector Function	Computes real-time cardinal directional bearings (e.g. `NE (42°)`) allowing dispatchers to give officers instant heading commands toward the victim.
Nominatim Geocoding	OpenStreetMap Reverse API	Converts raw WGS84 latitude/longitude coordinates into human-readable street addresses, campus buildings, and Kimberley residential quarters with memory caching.
IP-Based Geo Fallback	Multi-provider Geo Resolver	Provides automated server-side geographic fallback (via GeoJS/ipwhois) when desktop or Linux browsers fail to report hardware GPS coordinates.
OSRM Routing Gateway	Open Source Routing Machine	Calculates vehicular and pedestrian street turn-by-turn navigation paths from the responding officer's position directly to the emergency beacon.

4. Real-Time Telemetry & Transactional Communications Alerts & Messaging

SERVICE / LIBRARY	PROTOCOL & FORMAT	ROLE & PURPOSE IN UPBS ARCHITECTURE
Resend API `v2 Cloud API`	HTTPS RESTful Email Delivery	Enterprise transactional email delivery infrastructure used for instant user provisioning, automated welcome and credential setup emails, and password recovery.
CAD Long-Poll Stream `Custom JSON API`	Asynchronous Fetch / Polling	Synchronizes the Mobile MDT Officer Console and Admin Command Center with new emergency alerts, status updates (`DISPATCHED`, `EN_ROUTE`, `RESOLVED`), and active alerts every 3 seconds.
Web Audio API `HTML5 AudioContext`	In-Memory Sound Synthesis	Synthesizes audible emergency countdown beeps and distress sirens directly in browser memory without downloading external MP3/WAV audio files.
3-Second Abort Interlock `Client-Side Timer Interlock`	DOM Modal & State Machine	Provides a mandatory 3-second countdown buffer preventing accidental panic alerts. If the student cancels within 3 seconds, transmission is aborted; otherwise, CAD dispatch commits.

5. Frontend User Interface & Native Mobile App Presentation Layer

TECHNOLOGY	SPECIFICATION	ROLE & PURPOSE IN UPBS ARCHITECTURE
Semantic HTML5 & Modern CSS3	Glassmorphic Design System	Clean, responsive, high-contrast dark theme optimized for low-light campus emergency scenarios. Built with zero external CSS frameworks (no Bootstrap/Tailwind overhead), ensuring sub-second initial load times on low-bandwidth mobile networks.
Vanilla ES6+ JavaScript	Framework-Free Frontend	Powers the responsive 56px app bar, slide-down navigation drawer, category selector chips, live coordinate telemetry, and abort modals without React/Vue bundle overhead.
Android Native SDK `API 24 to 34 (Java 17)`	Native Companion App (APK)	Native Android shell (`za.ac.spu.upbs.student`) featuring hardware-accelerated WebView, permission management (`ACCESS_FINE_LOCATION`, `VIBRATE`), custom User-Agent filtering, and hardware back-button interlocks.
Tactile SOS Actuator	3D CSS Gradient & Ripples	High-impact 216px tactile dome button with pulsating sonar waves, multi-layer drop shadows, and active feedback designed to be pressed instantly under extreme stress.

6. Authentication, Cryptography & Role-Based Security Identity & Access Governance

SECURITY LAYER	MECHANISM	ROLE & PURPOSE IN UPBS ARCHITECTURE
Custom User Model	`accounts.User` (AbstractUser)	Replaces standard Django usernames with institutional identification numbers (e.g. `202474004`) and institutional email lookup support across all portals.
Password Hashing	PBKDF2 with SHA-256 (720k cycles)	NIST-compliant cryptographic password storage ensuring resistance against dictionary and brute-force GPU attacks.
Role-Based Access (RBAC)	4-Tier System Roles	Strict isolation across `STUDENT`, `STAFF`, `SECURITY`, and `ADMIN`. Prevents unauthorized elevation and redirects users to their designated portals.
Signed Cryptographic Tokens	Django `PasswordResetTokenGenerator`	Generates single-use HMAC-SHA256 tokens expiring in 24 hours for verified account activation and password recovery via email.

7. DevOps, Cloud Infrastructure & CI/CD Pipeline Hosting & Production Pipeline

SERVICE / TOOL	CONFIGURATION	ROLE & PURPOSE IN UPBS ARCHITECTURE
Render PaaS Cloud	Managed Web Service	Hosts the production web application with automated container deployments from GitHub, Let's Encrypt SSL/TLS certificates, and automated health checks.
Custom Domain & SSL	`https://youthalliance.co.za`	Custom institutional domain with automated CNAME redirection, HSTS enforcement, and strict-origin referrer headers.
Automated Build Pipeline	`build.sh` Shell Script	Automated CI/CD hook executing dependency installation, static compilation, database migrations, master admin provisioning, and test data cleanup on every commit.
Keepalive Heartbeat Daemon	`/healthz/` Probe Endpoint	Lightweight diagnostic route measuring live PostgreSQL round-trip latency and preventing free-tier container sleep during operational hours.
Django Test Suite	43 Unit & Integration Tests	Comprehensive test automation suite validating authentication, RBAC, alert lifecycles, geocoding fallbacks, and admin security before production release.